

PROJECT REPORT

Script-Controlled ACL – Restrict Record Access Based on Field Value

ServiceNow Project

Team ID: 6ab800af30f9767f87729b97

Team Members: ASWIN J C, Anton Viju J V, Sujith R, Swapnesh M G, Ragul A

Date: 29 September 2026

1. INTRODUCTION

1.1 Project Overview

The project demonstrates a script-controlled Access Control List (ACL) implementation in ServiceNow. The system uses roles and record conditions to control access to the custom Institution Details table. The intended access model permits the EEE user scenario to access the configured EEE records, while administrators retain full access.

1.2 Purpose

The purpose is to demonstrate how ServiceNow ACLs can protect records by evaluating user roles and configured field values. The project covers READ, CREATE, WRITE and DELETE access controls and validates both authorized and unauthorized access scenarios.

2. IDEATION PHASE

2.1 Problem Statement

Users of an institution record system may need access only to records relevant to their branch, while administrators require broader access. Without appropriate ACL rules, unauthorized users may be able to view or modify records that should be restricted.

2.2 Empathy Map Canvas

The empathy analysis considers users who need relevant institutional records, administrators who manage the complete dataset, and users who may be affected when access is either too

broad or too restrictive. Key areas considered are what users think and feel, hear, see, say and do, their pain points, and the gains expected from controlled access.

2.3 Brainstorming

The ideation process considered access-control approaches and prioritized a ServiceNow ACL solution that combines role-based authorization with record-level conditions. The selected idea was the Script-Controlled ACL approach because it directly demonstrates controlled access for the project table.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey follows the user from entering the ServiceNow environment, accessing the Institution Details area, attempting an operation, receiving an authorization decision, and continuing with the permitted workflow. Positive moments include appropriate access and clear administrative control; negative moments include denied access when a required permission is absent.

3.2 Solution Requirement

Functional requirements include authenticated access, role-based authorization, Branch-based READ control, CREATE authorization using bb2, WRITE authorization using bb3, DELETE authorization using bb4, and administrator access. Non-functional requirements include usability, security, reliability, performance, availability and scalability considerations.

3.3 Data Flow Diagram

The project flow begins with a user request to access Institution Details. ServiceNow evaluates the applicable ACL, user role and relevant record information, and then allows or denies the requested operation. Authorized requests proceed to the table operation; unauthorized requests are blocked.

DFD: Use the project-specific DFD from the separate DFD and User Stories document where the final report requires the diagram.

3.4 Technology Stack

Platform: ServiceNow. Core implementation: ServiceNow custom table, Access Control Lists, roles, ACL conditions and server-side ACL scripting. Custom table: u_institution_details. The project uses the ServiceNow platform's security and record-access mechanisms.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer segments include institutional users who need relevant records and administrators who require complete management access. The root problem is uncontrolled or overly broad record access. The proposed solution applies ACL rules so that access decisions are based on the user's assigned role and configured record conditions.

4.2 Proposed Solution

Create an Institution Details custom table and protect it with separate READ, CREATE, WRITE and DELETE ACLs. The READ ACL uses an advanced script and role/record logic, while CREATE, WRITE and DELETE are controlled through their corresponding roles. Administrators retain full access.

4.3 Solution Architecture

User → ServiceNow interface → ACL evaluation → role/condition validation → Institution Details table → allow or deny the requested operation. The architecture uses ServiceNow's built-in security and database mechanisms rather than an external application server.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was organized into sprints covering environment and user/table setup, ACL development, and testing/validation. The backlog included user-role setup, custom table and fields, READ ACL implementation, CREATE/WRITE/DELETE ACLs, and final validation and documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional testing covers READ, CREATE, WRITE and DELETE permissions, including authorized and unauthorized cases. Performance-oriented validation covers normal access response, repeated ACL evaluation, multi-record list access, and concurrent-user validation where supported by the test environment.

Actual response-time measurements and final Pass/Fail values must be entered from executed tests; this report does not invent performance measurements.

7. RESULTS

7.1 Output Screenshots

1. Institution Details table/list with test records and Branch values.
2. EEE user accessing the permitted records.
3. Unauthorized user showing restricted access.
4. Administrator accessing the records.
5. CREATE permission / New record screen.
6. WRITE permission / edited record.
7. DELETE permission / deletion result.
8. ACL configuration showing the relevant role, condition and script.

The screenshot shows the 'User - New Record' form in ServiceNow. The form is for creating a new user record. It includes fields for User ID (EEE User), First name (EEE), Last name (User), Title, Department, Email (eeeuser@gmail.com), Language (-- None --), Calendar integration (Outlook), Time zone (System (America/Los_Angeles)), Date format (System (2026-08-19)), Business phone, Mobile phone, and Photo (Click to add...). There are also checkboxes for Password needs reset, Locked out, Active (checked), and Internal Integration User. A 'Submit' button is at the bottom left.

servicenow

AllFavoritesHistoryWorkspaces

Role - bb1

Search

UpdateDelete

Name

bb1

Application

Global

Elevated privilege

Description

UpdateDelete

servicenow

AllFavoritesHistoryAdmin

Table - Institution Details

Search

DeleteUpdateDelete All Records

Table

Institution Details

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label

Institution Details

Application

Global

* Name

u_institution_details

Remote Table

ColumnsControlsApplication Access

servicenow

AllFavoritesHistoryAdmin

Table - Institution Details

Search

DeleteUpdateDelete All Records

Table

Institution Details

Table Columns

for text

Search

1 to 6 of 6

New

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
	Updated by	String	(empty)	40	false	false
	Updated	Date/Time	(empty)	40	false	false
	Updates	Integer	(empty)	40	false	false
	Created by	String	(empty)	40	false	false
	Sys ID	Sys ID (GUID)	(empty)	32	false	false
	Created	Date/Time	(empty)	40	false	false
	Student Roll Number	String				false
	Student Name	Reference	User			false
	Faculty Name	Reference	User			false
	Branch	Choice				false
	Email	String				false
	Phone Number	String				false
	Description	String				false
	Insert a new row...					

servicenow

AllFavoritesHistoryAdmin

Institution Details

Search

Actions on selected rows...

New

All

	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
	ECE			Abraham Lincoln		Abel Tuter	STU100
	CSE			Andrew Och		Amos Linnan	STU300
	EEE			Allyson Gillispie		Alejandra Prenatt	STU200

Search

System Administrator



System Administra...

System Administrator

ServiceNow

 Profile

 Preferences

 Keyboard shortcuts

 Impersonate user

 Elevate role

 Printer friendly version

 Log out

Student Name

Abel Tuter

Amos Linnan

Alejandra Prenatt

Abraham Lincoln

Abel Tuter

Elevate role

×

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

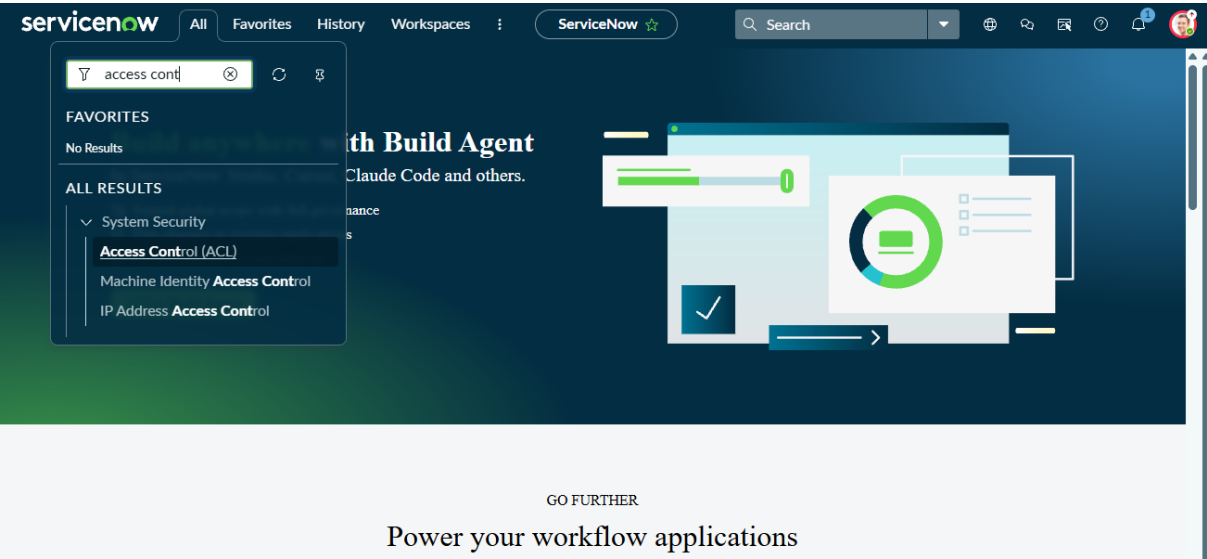
AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel

Update



ServiceNow interface showing the 'Access Controls' list. The table displays various access control records with columns: Name, Decision Type, Operation, Type, Active, Updated by, and Updated.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-va-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39
\$studio	Allow If	read	ui_page	true	admin	2016-01-09 16:21:38
\$upgrade_client	Allow If	read	ui_page	true	admin	2025-11-10 16:44:04
\$upgrade_monitor	Allow If	read	ui_page	true	admin	2025-09-29 22:40:08

ServiceNow interface showing the 'Access Control - New Record' form. The form includes fields for Name, Decision Type, Operation, Type, Application, Active, and Advanced. A warning message is displayed at the top: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.'

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record (info icon)

* Operation: read (search icon) (info icon)

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] (info icon)

Description: (text area)

Application: Global (info icon)

Active: ☒

Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

Data Condition

Condition

No. of records matching the condition: 3

Add Filter Condition

Add OR Clause

Branch

is

EEE

AND

OR

[Add Filter Condition](#) [Add OR Clause](#)

AND OR **X**

Advanced Condition

Advanced Condition

Controlled by References

Script ☒ Turn on ECMAScript 2021 (ES12) mode


```
1  (function () {  
7      if (gs.hasRole('bb1')){  
8          return true;  
9      }  
10     // Deny access for all others  
11     return false;  
12 })();  
13
```



```
1 (function () {  
7     if (gs.hasRole('bb1')){  
8         return true;  
9     }  
10    // Deny access for all others  
11    return false;  
12})();  
13
```

Access Control - u_institution_details

Search

UpdateDelete

* Type

record

Application

Global

* Operation

create

Active

☒

Admin overrides

☒

Advanced

☐

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

-- None --

Description

Allow create for records in u_institution_details, for users with role bb2.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

1 to 1 of 1

servicenow

AllFavoritesHistoryAdminAccess Controls

Search

UpdateDelete

Access Controlu_institution_details

* Type

record

* Operation

write

Admin overrides

☒

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Allow write for records in u_institution_details, for users with role bb3.

Application

Global

Active

☒

Advanced

☐

Access Control Configuration for u_institution_details

- * Type: record
- * Operation: delete
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: Allow delete for records in u_institution_details, for users with role bb4.

Conditions

Output Image:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

8. ADVANTAGES & DISADVANTAGES

Advantages: role-based access control; record-level restriction; separate permissions for READ/CREATE/WRITE/DELETE; administrator override; centralized ServiceNow security configuration; improved protection of institutional records.

Disadvantages: ACL configuration requires careful testing; incorrect roles or conditions can unintentionally deny or allow access; advanced scripts increase maintenance requirements; changes to roles or table fields may require ACL review.

9. CONCLUSION

The project demonstrates how ServiceNow ACLs can be used to control record access using roles and field-based conditions. The design separates the four major record operations and provides a practical security model for the Institution Details table.

10. FUTURE SCOPE

- Extend the model to additional branches and departments.
- Introduce more granular field-level security where required.
- Add audit and monitoring mechanisms for access-related events.
- Create reusable ACL patterns for additional custom tables.
- Expand automated regression testing for ACL changes.
- Integrate the access model with a larger institutional ServiceNow application.

11. APPENDIX

Source Code (if any)

ServiceNow ACL configuration and scripts used in the project. Include exported/update-set or relevant project source files in the GitHub repository where applicable.

```
(function () {  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    // Deny access for all others  
    return false;  
})();
```

Dataset Link

Not applicable as an external dataset is not required for this ServiceNow ACL project.

GitHub & Project Demo Link

GitHub Repository: <https://github.com/aswinjc/Script-Controlled-ACL-Restrict-Record-Access-Based-on-Field-Value>

Project Demo Video:

https://drive.google.com/file/d/17cr4pFSlvqO9uVhqpPKrUoeEZ2VmLo_E/view?usp=sharing